Greek labels : $0 \to 3$

Latin labels : $1 \to 3$

$$(v'^a \otimes u'^b) = R^a{}_c R^b{}_d \, v^c \otimes u^d$$

Asymmetry transformation S

$$V'^a = S^a{}_b V^b$$

$$S(T_2)\, S(T_1) = S(T_2 \circ T_1)$$

vectors that transform with S : upper indices vectors.

denotes matrix representation

Consider $S^{-1}(T_1)$

$$S^{-1}(T_2)\, S^{-1}(T_1) \neq S^{-1}(T_2 \circ T_1)$$

should be

$$S^{-1}(T_1)\, S^{-1}(T_2) = S^{-1}(T_2 \circ T_1)$$

$(AB)^T = B^T A^T$

$$\left(S^{-1}(T_1)\, S^{-1}(T_2)\right)^T = \left(S^{-1}(T_2 \circ T_1)\right)^T$$

$$(S^{-1})^T(T_2)\, (S^{-1})^T(T_1) = (S^{-1})^T (T_2 \circ T_1)$$

$$(S^{-1})^T = (S^T)^{-1} \equiv S^{-T}$$

vectors that transform with S^{-T} are lower indices vectors

$$V'_a = (S^{-T})_a{}^b V_b$$

compare w/ $S^a{}_b$!

$$\boxed{\begin{aligned} R^T &= R^{-1} \\ R^{-T} &= R \end{aligned}}$$

we didn't learn about 2 types of vectors because in ordinary notation $R^{-T} = R$.

However, in LT, that's not necessarily the case.

① $V'_a U'^a = (S^{-T})_a{}^b V_b (S)^a{}_c U^c$

index contraction

verify that partial derivatives transform w/ S^{-T}

$$= V_b U^c (S^{-T})_a{}^b S^a{}_c$$

$$= V_b U^c (S^{-1})^b{}_a S^a{}_c$$

$$= V_b U^c (S^{-1} S)^b{}_c$$

$$= V_b U^c \delta^b{}_c$$ → Kronecker delta

$$= V_b U^b$$

②

$t^{ab}, t^a{}_b, t_a{}^b, t_{ab}$

$\binom{2}{0}$ tensor $\binom{1}{1}$ tensor $\binom{0}{2}$ tensor

$t^{abc}{}_{de} \leftarrow \binom{3}{2}$ tensor.

$$t'^a{}_b = S^a{}_c (S^{-T})_b{}^d t^c{}_d$$

for example, consider $a, b \in \{0,1,2\}$.

$t^{ab}{}_a = \left(\sum_{a=0}^{2} t^{a0}{}_a, \sum_{a=0}^{2} t^{a1}{}_a, \sum_{a=0}^{2} t^{a2}{}_a\right)$

thus is a vector!

why is this a vector?

$$t'^{ab}{}_a = (S^a{}_c)(S^b{}_d)(S^{-T}{}_a{}^e)\, t^{cd}{}_e$$

$$= (S^a{}_c)(S^{-1})^e{}_a (S^b{}_d)\, t^{cd}{}_e$$

What is an example of this??

$$= \delta^e_c\, \delta^b_d\, t^{cd}{}_e$$

$$= \underline{\delta^b_d\, t^{cd}{}_c} = t^{cb}{}_c$$

$\overset{\text{oo}}{C}\omega_1 \quad \overset{\text{or}}{\omega_2} \quad \begin{bmatrix} I_1 & I_2 \\ I_2 & I_4 \end{bmatrix}$

$\vec{L} = \hat{I}\vec{\omega}$ → the moment of inertia is a tensor

$L^a = I^a{}_b \omega^b$

$\frac{1}{2}\vec{\omega}^T \hat{I} \vec{\omega}$

$\omega_a I^{ab} \omega_b$

try to work these indices out.

Here $(R^{-T}) = R$ so upper/lower indices X matter

$$I = \begin{pmatrix} I_{xx} & I_{xy} & I_{xz} \\ I_{yx} & I_{yy} & I_{yz} \\ I_{zx} & I_{zy} & I_{zz} \end{pmatrix} \qquad \omega = \begin{pmatrix} \omega_x \\ \omega_y \\ \omega_z \end{pmatrix}$$

$$\vec{\omega}^T I \vec{\omega} = \underbrace{\begin{pmatrix} \overset{00}{\omega_x} & \overset{01}{\omega_y} & \overset{02}{\omega_z} \end{pmatrix}}_{1\times 3} \underbrace{\begin{pmatrix} \overset{00}{I_{xx}} & \overset{01}{I_{xy}} & \overset{02}{I_{xz}} \\ \overset{10}{I_{yx}} & \overset{11}{I_{yy}} & \overset{12}{I_{yz}} \\ \overset{20}{I_{zx}} & \overset{21}{I_{zy}} & \overset{22}{I_{zz}} \end{pmatrix}}_{3\times 3} \underbrace{\begin{pmatrix} \overset{00}{\omega_x} \\ \overset{10}{\omega_y} \\ \overset{12}{\omega_z} \end{pmatrix}}_{3\times 1}$$

$$\underbrace{\left(\overset{00}{\omega_x} \overset{00}{I_{xx}} + \overset{01}{\omega_y} \overset{10}{I_{yx}} + \overset{02}{\omega_z} \overset{20}{I_{zx}} \right.}_{1\times 3}$$

$$\overset{01}{\omega_x I_{yx} + \omega_y I_{yy} + \omega_z I_{yz}}$$

$$\left. \overset{02}{\omega_x I_{zx} + \omega_y I_{zy} + \omega_z I_{zz}} \right)$$

$$\begin{pmatrix} \overset{00}{\omega_x} \\ \overset{10}{\omega_y} \\ \overset{20}{\omega_z} \end{pmatrix}$$

columns aligns with rows

$$\omega_{\underline{a}} I^{\underline{a}}{}_{\underline{b}} \omega^{b}$$

Tensor Symmetries

Suppose $t^{ab} = -t^{ba}$

↳ antisymmetric tensor

$$t'^{ab} = S^a{}_m S^b{}_n t^{mn}$$
$$= -S^a{}_m S^b{}_n t^{nm}$$
$$= -t^{ba}$$

Antisymmetry of tensors are frame independent.

$$\vec{p} = \begin{pmatrix} p^1 \\ p^2 \\ p^3 \end{pmatrix}$$

t^{ab} $\quad a,b = 1,2,3$ → 9 indices

you only need to know the upper triangle to know the whole tensor.

Suppose $t^{ab} = -t^{ba}$

then $t^{11} = t^{22} = t^{33} = 0$

3 independent components.

$$\begin{cases} t^{12} = -t^{21} & t^{23} = -t^{32} \\ t^{13} = -t^{31} \end{cases}$$

$4 \times 4 \times 4 \times 4 = 256.$

$R^{\mu\nu\rho\sigma}$

$\mu, \nu, \rho, \sigma = 0,1,2,3$ [256 comp. without symmetry]

$$\begin{cases} R^{\mu\nu\rho\sigma} = -R^{\nu\mu\rho\sigma} \\ R^{\mu\nu\rho\sigma} = -R^{\mu\nu\sigma\rho} \\ R^{\mu\nu\rho\sigma} = R^{\rho\sigma\mu\nu} \end{cases}$$

$$\frac{d^2(d^2-1)}{12}$$

indices for the Riemann curvature tensor.

$d = 4$

4

$\frac{4 \times 16 \times 15}{3 \cdot 12} = 20$ components!

Invariant Tensors.

① $\delta^a{}_b = \begin{cases} 1, & a=b \\ 0, & a\neq b. \end{cases}$

$$\delta'^a{}_b = S^a{}_c \left(S^{-T}\right)_b{}^d \, \delta^c{}_d$$

$$= S^a{}_c \left(S^{-T}\right)_b{}^c$$

$$= \left(S^{-T}\right)_b{}^c \left(S^T\right)_c{}^a$$

$$= \left(S^{-T} S\right)_b{}^a = \delta_b{}^a = \delta^a{}_b$$

② there are 2 sym. tensors depending on S itself.